MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Section 1.3: Matrix-vector product and linear maps

Matrix-vector product

A vector can be written as a row or as a column. When a matrix acts on a vector in this course, we usually write the vector as a column:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

From now on, when we write $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{x}$ is interpreted as a column vector.
If A is an $m \times n$ matrix, then A has n columns. We write

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

where each $\mathbf{a}_j$ is a column vector in $\mathbb{R}^m$.

Definition: Matrix-vector product

Let $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$ be an $m \times n$ matrix and let $\mathbf{x} = (x_1, \dots, x_n)$ be a column vector in $\mathbb{R}^n$. The

$\boxed{\qquad\qquad\qquad\qquad}$ is

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Activity: Computing a matrix-vector product

Compute $A\mathbf{x}$ where

$$A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.$$

Theorem: Properties of matrix-vector products

Let A and B be $m \times n$ matrices, let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and let r be a scalar. Then

- $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$,
- $A(r\mathbf{x}) = r(A\mathbf{x}) = (rA)\mathbf{x}$,
- $(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}$.

Why is this true?

We prove the first property. Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, $\mathbf{x} = (x_1, \dots, x_n)$, and $\mathbf{y} = (y_1, \dots, y_n)$. By the definition above,

$$\begin{aligned} A(\mathbf{x} + \mathbf{y}) &= (x_1 + y_1)\mathbf{a}_1 + \cdots + (x_n + y_n)\mathbf{a}_n \\ &= A\mathbf{x} + A\mathbf{y}. \end{aligned}$$

Theorem: Row dot-product view

If A is an $m \times n$ matrix with rows $\mathrm{row}_1(A), \dots, \mathrm{row}_m(A)$ as n-vectors, then

$$
A\mathbf{x} = \begin{bmatrix} \mathrm{row}_1(A)\cdot\mathbf{x} \\ \mathrm{row}_2(A)\cdot\mathbf{x} \\ \vdots \\ \mathrm{row}_m(A)\cdot\mathbf{x} \end{bmatrix}.
$$

Why is this true?

Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, where $\mathbf{a}_j = (a_{1j}, \dots, a_{mj})$, and let $\mathbf{x} = (x_1, \dots, x_n)$. Since

$$
A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n,
$$

its i-th entry is

$$
x_1 a_{i1} + \cdots + x_n a_{in} = \mathrm{row}_i(A)\cdot\mathbf{x}.
$$

Stacking these entries gives the stated formula.

Example: Computing the same product using rows

We revisit Computing a matrix-vector product using row dot products. Compute Ax where

$$
A = \begin{bmatrix} -1 & 4 & -5 \\ 3 & 1 & -2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}.
$$

Note: Shape habit for matrix-vector products

Before forming Ax, check that the number of columns of A equals the number of entries of $\mathbf{x}$. If A is $m \times n$ and $\mathbf{x} \in \mathbb{R}^n$, then $A\mathbf{x} \in \mathbb{R}^m$: its m entries come from the m row dot products above.

Note: Rows measure; columns contribute

A matrix-vector product has two complementary readings. In the row view, each row measures the input by taking a dot product with $\mathbf{x}$. In the column view,

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n,$$

so the input coordinates tell how much each column contributes to the output.

Examples of matrix-vector products

The same product $A\mathbf{x}$ can represent different operations, depending on what the rows and columns of A mean. The next activities show three common patterns: scoring, selecting, and measuring changes.

Activity: Scoring objects by features

Let the rows of X represent three objects with two features:

$$X = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 2 \end{bmatrix}, \qquad \mathbf{w} = (2, -1).$$

Compute $X\mathbf{w}$. Interpret the result.

Activity: A selector matrix

Let

$$S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}.$$

Compute $S\mathbf{x}$. What does S do?

Activity: A difference matrix

Let

$$D = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}.$$

Compute $D\mathbf{x}$. What does D measure?

Activity: Document ranking in matrix code

In Word-count vectors, we ranked documents by cosine similarity one score at a time. The following code repeats that ranking using a data matrix whose rows are the document vectors.

```
import numpy as np

doc_names = np.array(["D1", "D2", "D3"])

X = np.array([
    [3., 0., 3.],
    [1., 1., 0.],
    [0., 2., 0.],
])

q = np.array([1., 0., 1.])

scores = (X @ q) / (
    np.linalg.norm(X, axis=1) * np.linalg.norm(q)
)
ranking = doc_names[np.argsort(scores)[::-1]]

scores, ranking
```

Output:

```
(array([1. , 0.5, 0. ]),
 array(['D1', 'D2', 'D3']))
```

1. What does the second row of `X` represent?
2. Which expression computes the three dot products $\mathbf{D}_i \cdot \mathbf{q}$?
3. What does `np.argsort(scores)[::-1]` do?

Example: The attention activity in matrix form

In Gathering context for the bird position, we computed the scores

$$\mathbf{q} \cdot \mathbf{k}_{\text{small}}, \qquad \mathbf{q} \cdot \mathbf{k}_{\text{red}}, \qquad \mathbf{q} \cdot \mathbf{k}_{\text{bird}}$$

one at a time. Matrix-vector multiplication computes the same scores at once.

Put the key vectors as the rows of

$$K = \begin{bmatrix} \mathbf{k}_{\text{small}}^T \\ \mathbf{k}_{\text{red}}^T \\ \mathbf{k}_{\text{bird}}^T \end{bmatrix}.$$

Then

$$K\mathbf{q} = \begin{bmatrix} \mathbf{k}_{\text{small}} \cdot \mathbf{q} \\ \mathbf{k}_{\text{red}} \cdot \mathbf{q} \\ \mathbf{k}_{\text{bird}} \cdot \mathbf{q} \end{bmatrix}.$$

For the numbers in Gathering context for the bird position,

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{q} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so

$$K\mathbf{q} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Activity: The same attention calculation in code

The following code performs the calculation from the token attention activity.

```
import numpy as np

tokens = np.array(["small", "red", "bird"])

q = np.array([1., 1.])

K = np.array([
    [1., 0.],
    [0., 1.],
    [1., 1.],
])

V = np.array([
    [4., 0.],
    [0., 4.],
    [4., 4.],
])

s = K @ q
alpha = s / s.sum()
out = alpha @ V

s, alpha, out
```

Output:

```
(array([1., 1., 2.]),
 array([0.25, 0.25, 0.5 ]),
 array([3., 3.]))
```

1. Which line computes the token scores?

Activity: The same attention calculation in code (continued)

2. Which token receives the largest weight?
3. Which line forms the weighted average?
4. Why do the entries of `alpha` add to 1?

Geometric matrix actions in $\mathbb{R}^2$

Definition: Identity matrix

The ______ I_n is the $n \times n$ matrix with 1s on the main diagonal and zeros elsewhere.

Fact

For every column vector $\mathbf{x} \in \mathbb{R}^n$, $I_n\mathbf{x} = \mathbf{x}$.

Definition: Standard basis

The j-th ______ $\mathbf{e}_j$ is the column vector with a 1 in position j and zeros elsewhere.

The term "basis" will be explained later in the course. For now, use the concrete description of $\mathbf{e}_j$ above.

Fact

If $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}$, then $A\mathbf{e}_j = \mathbf{a}_j$. Applying a matrix to a standard basis vector picks out a column.

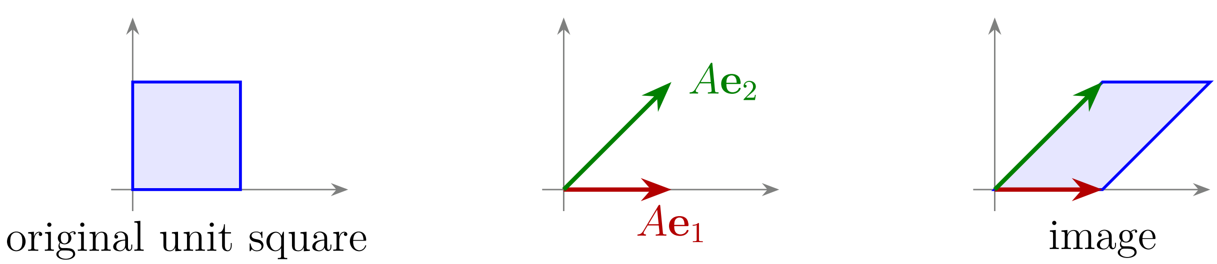
In $\mathbb{R}^2$, a matrix $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix}$ sends $\mathbf{e}_1$ to $\mathbf{a}_1$ and $\mathbf{e}_2$ to $\mathbf{a}_2$. Since every vector $\mathbf{x} = (x_1, x_2)$ can be written as $x_1\mathbf{e}_1 + x_2\mathbf{e}_2$, we have

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2.$$

Thus a 2×2 matrix is geometrically determined by where it sends the two coordinate directions.

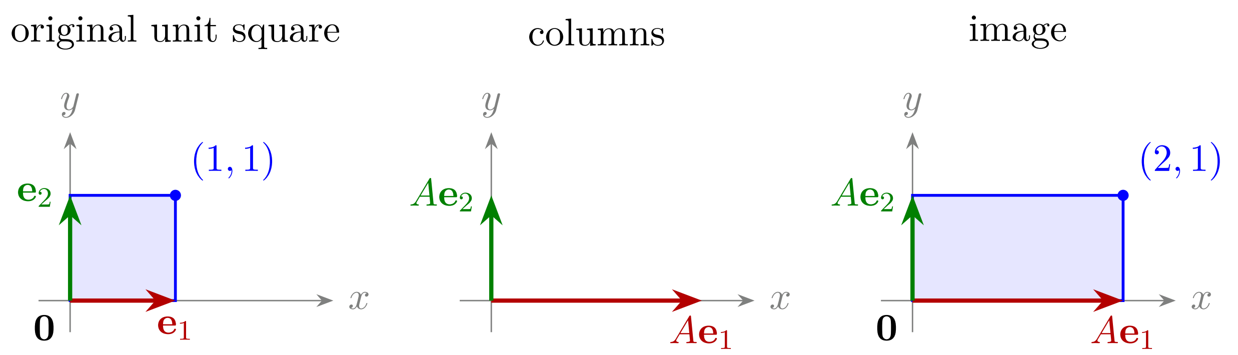
To understand a 2×2 matrix A, apply it to the four corners of the unit square. The image is usually a parallelogram. The two edges leaving the origin are the columns of A. We will refer to this as the

.



$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Figure. A 2×2 matrix sends the unit square to the parallelogram determined by its two columns. This is the geometric version of columns contribute.



$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Figure. A horizontal stretch by a factor of 2 sends the unit square to a rectangle twice as wide. The first column doubles the horizontal basis vector, while the second column leaves the vertical basis vector unchanged.

- **Matrix:** $\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$
  $A\mathbf{e}_1$: $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$
  $A\mathbf{e}_2$: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  Name: horizontal stretch
  Unit-square effect: The right edge moves farther right.

- **Matrix:** $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
  $A\mathbf{e}_1$: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
  $A\mathbf{e}_2$: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  Name: projection onto the y-axis
  Unit-square effect: Horizontal information is forgotten.

- **Matrix:** $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
  $A\mathbf{e}_1$: $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

Ae_2: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
Name: reflection across the y-axis
Unit-square effect: The square flips left.

- **Matrix:** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  Ae_2: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
  Name: reflection across the line $y = x$
  Unit-square effect: The square stays in the same region, but the horizontal and vertical coordinate directions exchange.

- **Matrix:** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
  Ae_2: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
  Name: horizontal shear
  Unit-square effect: The bottom edge stays fixed; the top edge shifts right.

- **Matrix:** $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
  Ae_1: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
  Ae_2: $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$
  Name: 90° counterclockwise rotation
  Unit-square effect: The square rotates around the origin.

- **Matrix:** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Ae_1: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
Ae_2: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
Name: projection onto the x-axis
Unit-square effect: Vertical information is forgotten.

For more illustrations of these transformations, see Lab U1: Vectors, Similarity, Attention, and Matrix Actions.

Activity: Horizontal shear

Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Compute $A\mathbf{e}_1$, $A\mathbf{e}_2$, and $A\mathbf{x}$, where $\mathbf{x} = (1, 1)$.

Note

For the shear matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, the row view gives

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (1, 1) \cdot (x, y) \\ (0, 1) \cdot (x, y) \end{bmatrix} = \begin{bmatrix} x + y \\ y \end{bmatrix}.$$

The first row measures horizontal plus vertical, while the second row measures vertical only. The column view gives

$$A \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

so the y-coordinate contributes both upward and rightward motion.

Activity: Reflection across the y-axis

Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Compute $A\mathbf{e}_1$, $A\mathbf{e}_2$, $A\mathbf{x}$, and $A\mathbf{y}$, where $\mathbf{x} = (1, 1)$ and $\mathbf{y} = (0, 1)$.

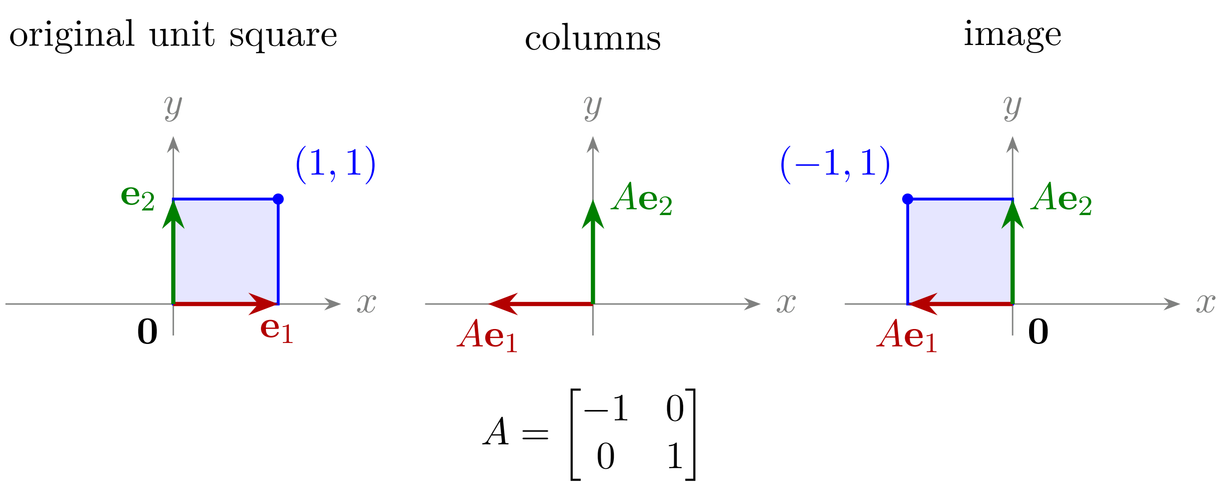


Figure. Reflection across the y-axis sends the unit square to its left-hand mirror image. The first column reverses the horizontal direction, while the second column leaves the vertical direction unchanged.

Activity: Reflection across the line y=x

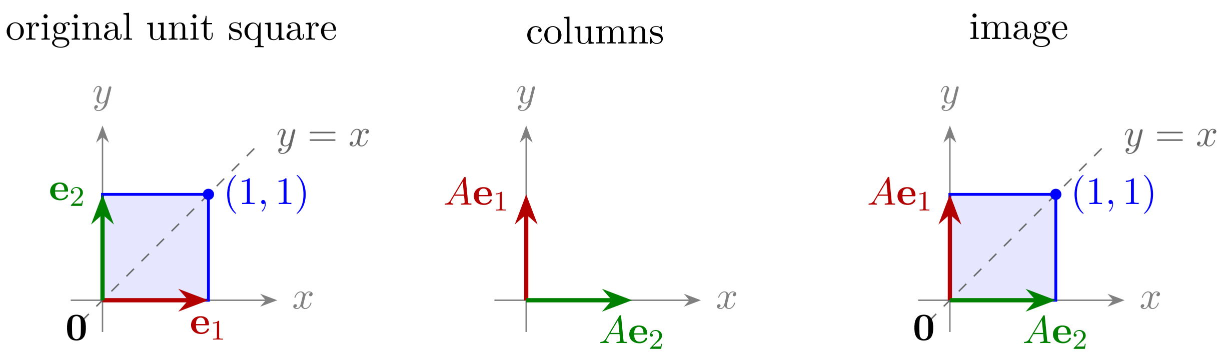
Let

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute $A\mathbf{e}_1$ and $A\mathbf{e}_2$.
2. Compute $A \begin{bmatrix} x \\ y \end{bmatrix}$.

Activity: Reflection across the line y=x (continued)

3. What happens to a point on the line $y = x$?



$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Figure. Three-panel unit-square visualization of reflection across y equals x, with exchanged basis arrows.

Linear and affine maps

A function assigns one output to each input. In this course, many functions have vectors as inputs and vectors as outputs:

$$F : \mathbb{R}^n \to \mathbb{R}^m.$$

The of F is its set of allowed inputs, $\mathbb{R}^n$, and its is the specified set containing its outputs, $\mathbb{R}^m$. A matrix gives one important source of such functions by the rule $\mathbf{x} \mapsto A\mathbf{x}$.

Definition: Matrix map

Given an $m \times n$ matrix A, the induced by A is the function $T_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by

$$T_A(\mathbf{x}) = A\mathbf{x}.$$

Definition: Linear map

A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is if

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T(\mathbf{u}) + T(\mathbf{v}), \\ T(c\mathbf{v}) &= cT(\mathbf{v}) \end{aligned}$$

for all vectors $\mathbf{u}, \mathbf{v}$ and all scalars c.

Theorem: Matrix maps are linear

If A is an $m \times n$ matrix, then the function

$$T_A : \mathbb{R}^n \to \mathbb{R}^m, \qquad T_A(\mathbf{x}) = A\mathbf{x}$$

is linear.

Why is this true?

Matrix-vector multiplication distributes over vector addition and scalar multiplication:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}, \qquad A(c\mathbf{v}) = cA\mathbf{v}.$$

These are exactly the two linearity rules.

Theorem: Every linear map has a matrix

Every linear map $T : \mathbb{R}^n \to \mathbb{R}^m$ is given by multiplication by a unique matrix. The columns of that matrix are $T(\mathbf{e}_1), \dots, T(\mathbf{e}_n)$.

Why is this true?

Write $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$ and let $A = [\,T(\mathbf{e}_1)\ \cdots\ T(\mathbf{e}_n)\,]$. By linearity and the column view of matrix-vector multiplication,

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n) = A\mathbf{x}.$$

This matrix is unique because its j-th column must be $A\mathbf{e}_j = T(\mathbf{e}_j)$.

Activity: Images of basis vectors determine the matrix

Suppose a linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ satisfies $T(\mathbf{e}_1) = (2,1)$ and $T(\mathbf{e}_2) = (-1,3)$. Find the matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

Warning

We use "linear map" and "linear transformation" interchangeably; "transformation" is often used when emphasizing geometry.

Definition: Affine map

A function of the form

$$\mathbf{x} \mapsto A\mathbf{x} + \mathbf{b}$$

is called ______. It is built from a linear map followed by a ______.

An affine map is linear only when $\mathbf{b} = \mathbf{0}$. A quick test is the zero vector: every linear map sends $\mathbf{0}$ to $\mathbf{0}$, but

$$A\mathbf{0} + \mathbf{b} = \mathbf{b}.$$

If $\mathbf{b} \neq \mathbf{0}$, the map is not linear.

Example: An affine layer

A layer in a neural network can take the form

$$\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}.$$

The matrix W mixes the input coordinates. The vector $\mathbf{b}$ shifts the result. Since

$$W\mathbf{0} + \mathbf{b} = \mathbf{b},$$

this map is affine. It is linear exactly when $\mathbf{b} = \mathbf{0}$.